SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 9: 9. Travelling Salesman Problem

Aim: To find a minimum-cost route that visits every city once and returns to the starting city.

Python Program

```
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]
n = len(graph)
cities = range(1, n)
min_cost = float('inf')
best_path = None

for p in permutations(cities):
    path = (0,) + p + (0,)
    cost = sum(graph[path[i]][path[i+1]] for i in range(n))
    if cost < min_cost:
        min_cost = cost
        best_path = path

print("Best path:", best_path)
print("Minimum cost:", min_cost)
```

Sample Output

```
Best path: (0, 1, 3, 2, 0)
Minimum cost: 80
```

Result: The program demonstrates the stated Artificial Intelligence problem.